NATIONAL INNOVATORS BATTLE (NIB) 2026

ROBORUGBY – PARTICIPANT GUIDELINES, RULES & REGULATIONS

Competition: RoboRugby
Theme: Learn • Build • Innovate • Compete
Registration Deadline: 15 December 2026
Grand Finale: 20 December 2026
Venue: National Innovators Battle (NIB) Arena Hall, Lucknow, Uttar Pradesh
Official Website: www.nibindia.org

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1. COMPETITION STRUCTURE

The RoboRugby competition will be conducted in two stages.

Round 1 – School-Level Competition

- Teams will participate in a live RoboRugby project presentation and demonstration at their respective school events.
- Participants must explain their robot's design, construction, control mechanism, and functionality.
- Shortlisted teams will qualify for the National Grand Finale.

Round 2 – National Grand Finale

- Shortlisted teams will compete with their working RoboRugby robots at the NIB Arena Hall.
- Teams will participate in competitive matches according to the tournament schedule.
- The winners will be determined based on match scores and competition performance.

2. TEAM REGISTRATION & GENERAL RULES

1. Every team must register through the official NIB website or the registration link provided by their school.
2. A unique Team ID will be allotted after successful registration.
3. Teams must mention their Team ID in all official communication.

4. Only registered participants are permitted to operate their team's robot during matches.
5. Participants must report at the venue according to the schedule shared by the organizers.
6. Each team is responsible for bringing its robot, controller, batteries, chargers, tools, and other necessary equipment.
7. Teams must follow the instructions of referees, technical inspectors, and organizers.
8. The decisions of the referee and organizing committee shall be final and binding.
9. Participants are advised to regularly check their registered email accounts, including spam and promotions folders, for official updates.
10. NIB shall not be responsible for late, lost, or misdirected registration submissions.

3. ROBOT SPECIFICATIONS

A. Dimensions & Weight

1. Maximum robot dimensions: **550 mm × 550 mm × 650 mm (Length × Breadth × Height)**.
2. Maximum permitted robot weight: **20 kg**.
3. Dimensions and weight include batteries, weapons, controllers mounted on the robot, and all other onboard components.
4. Every team must declare all robots and major weapon systems before its first match.
5. No major modifications to the declared robot or weapon system will be permitted without approval from the technical inspection committee.
6. Robots exceeding the specified dimensions or weight will not be allowed to participate until they meet the requirements.

B. Robot Mobility

1. Robots must have controlled and clearly visible ground mobility.
2. Both wired and wireless robots are permitted.
3. Flying robots or robots designed to leave the arena are not permitted.
4. Robots must be capable of controlled movement in the arena.
5. A robot that becomes completely non-functional or cannot move during a match may be declared inactive by the referee.
6. Both teams must pass the mobility test before entering their respective matches.

C. Wired & Wireless Control

1. Teams may use wired or wireless control systems.
2. Wireless robots must use reliable communication modules that do not interfere with other teams' robots.
3. Teams using wireless communication must be capable of changing their operating frequency/channel if instructed by the technical committee.
4. Failure to resolve frequency interference may result in disqualification.
5. Wired robot control cables must be maintained at least 1 metre above the ground to avoid interference with the arena and other robots.
6. Wireless systems must comply with applicable radio-frequency regulations.
7. The use of radio jamming or intentional interference with another robot's control system is strictly prohibited.

4. ROBO RUGBY – MATCH RULES

1. The objective is to score maximum points by controlling, pushing, striking, or moving the opponent's robot and successfully achieving goals.
2. Teams may use pushing, kicking, lifting, flipping, and other permitted contact mechanisms.
3. Robot-to-robot combat is permitted within the approved weapon and safety regulations.
4. A match will begin only after the referee gives the official start signal.
5. Robots must remain stationary during the countdown: 3, 2, 1.
6. No team is allowed to touch, adjust, or interfere with its robot during an active match without the referee's permission.
7. Teams may request a timeout, but approval will be entirely at the referee's discretion.
8. A break of approximately 1–2 minutes may be provided between rounds or match periods, as decided by the referee.
9. Compensation time may be granted at the referee's discretion in case of technical interruptions.
10. A robot that becomes completely inactive may be declared a dead robot. The opposing team may be declared the winner according to the match rules.
11. Both robots will undergo a movement check before every match.

12. Teams may raise objections or claims through the referee, but must continue playing until the referee officially stops the match.
13. Teams must immediately stop their robots when the referee blows the whistle or announces the end of the match.

5. SCORING SYSTEM

The following scoring system will be used during RoboRugby matches:

Action	Points
Successful Hit / Touch / Contact	20
Opponent Robot Toppled or Pushed Out of Arena	50
Successful Goal	100

Scoring Guidelines:

1. Points will be awarded only when the referee confirms a valid action.
2. Repeated contact will be scored according to the referee's interpretation of valid hits and the official scoring procedure.
3. A successful goal must meet the arena's designated goal requirements.
4. A robot leaving the arena due to a valid opponent action will be considered according to the 50-point scoring rule.
5. The team scoring the highest total points at the end of the match will be declared the winner.
6. In case of a tie, the referee may announce an additional round or tie-breaker.
7. The referee's scoring decision will be final.

6. ALLOWED ACTIONS

Subject to weapon approval and safety regulations, the following actions are permitted:

1. Pushing and striking the opponent's robot.
2. Kicking and thrashing using approved mechanisms.
3. Lifting and flipping the opponent's robot.
4. Using approved spinning, cutting, magnetic, and mechanical mechanisms.
5. Attempting to push the opponent out of the arena.

6. Using creative mechanical designs to gain a competitive advantage.

All actions must remain within the approved safety boundaries.

7. WEAPONS & MECHANICAL SYSTEMS

Robots may use approved mechanical weapons and mechanisms such as:

- Flippers and lifting mechanisms.
- Spinning hammers.
- Guarded cutters and saw mechanisms.
- Mechanical pushers and strikers.
- Approved magnetic mechanisms.
- Other mechanical systems approved by the technical inspection committee.

Pneumatic & Hydraulic Systems

1. Pneumatic and hydraulic systems may be permitted subject to prior approval and safety inspection.
2. Pneumatic systems may use only approved, non-flammable gases.
3. Maximum permitted pneumatic outlet pressure: **8 bar**.
4. Hydraulic systems must use approved non-flammable liquids.
5. Pressure vessels, cylinders, hoses, valves, and connections must be properly rated and securely mounted.
6. All pressurized systems must have suitable pressure-relief and emergency-release arrangements.
7. Any system considered unsafe by the technical committee will not be permitted.

Strictly Prohibited Weapons & Mechanisms

The following are strictly prohibited:

1. Flame-based or fire-producing weapons.
2. Explosives or explosive mechanisms.
3. Flammable liquids or chemical-based weapons.
4. Liquid projectiles.
5. Tethered or untethered projectiles.

6. High-power magnets or electromagnets capable of creating unsafe interference or hazards.
7. Radio jamming devices.
8. Tasers, Tesla coils, or high-voltage shock devices.
9. Nets, tapes, strings, or entanglement mechanisms designed to immobilize robots.
10. Any weapon designed to damage the arena or endanger participants, judges, or spectators.
11. Any mechanism that intentionally creates a hazard outside the competition arena.

Note: All weapons and moving mechanisms must pass technical inspection before being used in competition. The organizers reserve the right to prohibit any mechanism considered unsafe.

8. BATTERY & POWER REGULATIONS

1. Each team must bring its own batteries and power equipment.
2. Only sealed, non-spillable batteries are permitted.
3. Maximum permitted operating voltage: **18V DC** between any two electrical points.
4. Battery voltage must be declared during technical inspection.
5. Batteries must be securely mounted and protected against impact, short circuits, and accidental disconnection.
6. Exposed wires, damaged batteries, and unsafe electrical connections are prohibited.
7. Battery charging must take place only in designated areas, if provided by the organizers.
8. Teams must bring their own chargers and compatible power accessories.
9. All batteries and power systems are subject to inspection before every match.

9. SPRING, PINNING & ENERGY-STORAGE SYSTEMS

1. Large springs used for driving or weapon mechanisms must have a safe and controlled loading and release mechanism.
2. Spring-loaded systems must not be manually activated while the robot is being handled in an unsafe condition.
3. Flywheels and other kinetic-energy storage devices must remain inactive during transportation and handling.
4. Such devices may be activated only in designated testing areas or inside the competition arena, with referee permission.

5. All stored-energy mechanisms must have suitable safety guards and emergency shutdown arrangements.

10. WARNINGS & PENALTIES

Warnings

The following actions may result in a warning:

1. Moving the robot before the official start signal or during the countdown.
2. Touching or interfering with a robot during an active match without referee permission.
3. Ignoring referee instructions.
4. Violating match procedures or creating unnecessary interruptions.
5. Any unsportsmanlike conduct during the competition.

Penalties

1. A second warning for a similar or repeated violation may result in a **20-point penalty**.
2. The referee may award additional penalties depending on the seriousness of the violation.
3. Both striker and defender robots must have been declared and approved for the match.
4. Intentionally damaging the arena will result in immediate disqualification.
5. Cutting or deliberately damaging the opponent's control wires is strictly prohibited and may result in disqualification.
6. Deliberate interference with another team's robot or control system may result in disqualification.
7. Repeated violations of safety instructions may lead to removal from the competition.

11. ARENA SPECIFICATIONS

1. Approximate arena dimensions: **8 ft × 8 ft**.
2. The arena may include hurdles, obstacles, goals, and other designated playing structures.
3. The arena layout and goal positions will be explained to participants before the competition.
4. Participants must not intentionally damage or modify the arena.
5. No participant is allowed to enter the arena during an active match.

6. Only authorized referees and technical officials may enter the arena during match interruptions.
7. Teams must operate their robots only within the designated competition area.

12. TECHNICAL INSPECTION & ROBOT DECLARATION

1. Every robot must pass technical inspection before its first match.
2. The inspection committee will verify dimensions, weight, voltage, mobility, weapon safety, and wireless communication.
3. Teams must declare all robots, batteries, and weapon mechanisms before their first match.
4. Any modification after inspection must be reported to the technical committee.
5. Judges may conduct additional inspections at any time during the competition.
6. Robots that fail safety or technical inspection will not be allowed to compete until the issue is resolved.
7. The technical committee may reject any robot or mechanism that presents a safety risk.

13. MATCH PROCEDURE

1. Teams will be called according to the official match schedule.
2. Both teams must report to the arena with their inspected robots.
3. The referee will verify robot readiness before starting the match.
4. The match will begin with a 3, 2, 1 countdown followed by the start signal.
5. Participants must operate their robots from their designated control positions.
6. The referee may pause or stop a match in case of safety concerns, technical problems, or rule violations.
7. At the end of the match, all robots must be switched off before being removed from the arena.
8. The referee will announce the final score and winning team.

14. DISPUTES & DECISION-MAKING

1. Any team wishing to raise an objection must approach the referee respectfully.
2. Only the registered team members may raise official match-related objections.

3. The referee may consult the technical or organizing committee when necessary.
4. Decisions regarding scoring, warnings, penalties, and disqualification shall be made by the authorized officials.
5. The decision of the referee and organizing committee shall be final and binding.

15. SAFETY & CODE OF CONDUCT

- Participants must wear appropriate footwear and follow all safety instructions.
- No participant may operate a robot outside designated testing or competition areas.
- Robots must be switched off before being carried, repaired, or inspected.
- Weapon systems must be mechanically secured during transportation.
- Participants must not activate spinning weapons, pneumatic systems, or other dangerous mechanisms outside authorized areas.
- Coaches and accompanying persons must not interfere with matches or attempt to influence referees.
- Any behaviour that threatens the safety or discipline of the event may result in disqualification.

16. AWARDS & RECOGNITION

Winners will be recognized based on their performance in the RoboRugby competition.

- First Position – Winner
- Second Position – Runner-Up
- Third Position – Second Runner-Up

Additional special awards may be announced by the organizers based on competition performance and merit.

17. OFFICIAL COMMUNICATION

All official announcements, competition updates, schedules, and changes to rules will be communicated through NIB's official website and registered email addresses.

Official Website: www.nibindia.org
Email: nationalinnovatorsbattle@gmail.com

A MESSAGE TO ALL ROBO RUGBY PARTICIPANTS

Build your robot, test your strategy, and prepare for the challenge!

RoboRugby is not just about winning a match. It is about engineering, innovation, teamwork, creativity, and learning through competition.

Respect your opponents, follow the rules, compete fairly, and let your robot demonstrate your engineering skills.

LEARN • BUILD • INNOVATE • COMPETE

TEAM NIB 2026
National Innovators Battle
Vigyan Pathshala & TeamTalks